A PBL-I Synopsis on

Early Dyslexia Identification and Confident Reading – ReadBright App

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1. INTRODUCTION TO PROBLEM

1.1 Understanding Dyslexia: A Neurological Difference

Dyslexia is a **common neurological difference** that primarily affects the ability to **read, write, and spell**, despite the individual having **normal or above-average intelligence**.

- **Not a Disease:** It is not a disease, but a difference in the brain's structure and function.
- **Core Challenge:** It particularly impacts **phonological processing**—the ability to recognize and manipulate the sounds of language. The core difficulty is connecting sounds to letters (e.g., C /k/, A /a/, T /t/ to form CAT).

1.2 The Critical Need for Early Identification

The importance of **early identification** cannot be overstated.

- When dyslexia goes unnoticed in early childhood, children often struggle quietly, leading to **long-term academic challenges, low self-esteem, and significant emotional stress.**
- **Global Prevalence:** Globally, an estimated **17% of students** have some degree of dyslexia.
- **India's High Rate:** The prevalence in India is notably high, impacting an estimated **15–20% of the student population**

1.3 Barriers to Dyslexia Management

The current landscape of dyslexia management is filled with **unseen barriers**:

- **Lack of Awareness:** There is a general lack of awareness among parents and educators, causing early signs to be **missed or misinterpreted.**
- **Cost Barrier:** Professional, specialized diagnostic assessments are often **prohibitively expensive and inaccessible**, creating a significant **socioeconomic**
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- **barrier** to receiving help.
- **Emotional Toll:** This lack of support often results in children being unfairly labeled as **lazy or unmotivated**, which compounds their emotional distress.
- **Inadequate Systems:** Existing support systems are insufficient, offering **limited specific guidance** for families and **insufficient specialized teacher training.**

1.4 Introducing ReadBright: The Digital Solution

This project, **ReadBright**, is an **Application-based solution** designed to address this fragmentation and inaccessibility.

- **Goal:** It aims to provide an **accessible, accurate, and continuous developmental tool** for early dyslexia identification and personalized intervention.
- **Outcome:** This approach fosters **confident reading and academic success** for young learners.

2. LITERATURE REVIEW

The proposed solution, **ReadBright**, is grounded in established educational psychology and leverages recent advancements in Artificial Intelligence (AI) and digital technology. This survey reviews the strengths and weaknesses of foundational methods, digital screening tools, and modern assistive technologies, drawing on recent, credible sources

Paper/Author/Year	Major Goals	Accuracy Parameter	Research Gaps
1. (2021) The Efficacy of Orton-Gillingham Approaches	To conduct a meta-analysis on the effectiveness of O-G approaches for improving reading skills in individuals with dyslexia.	Effect Size (ES) = 0.22 (Small to Moderate)	High cost and time commitment; lack of scalability for widespread, low-cost implementation in general education settings.
2. (2024) Automated Dyslexia Screening Using Children's Handwriting	To propose a method for predicting dyslexia risk by analyzing children's handwriting features using machine learning.	Accuracy up to 98% (Artificial Neural Networks - ANN)	Requires specialized hardware (digital pen/tablet); limited to handwriting analysis, potentially missing other phonological deficits.

3. (2025) Multimodal Analysis for Early Detection	To develop a comprehensive diagnostic method combining linguistic and behavioral markers (speech, eye-tracking, writing).	F1-score = 0.93 (High predictive power)	Requires specialized hardware and complex data processing; still in research stage, not a practical low-cost screening tool.

3.COMPARITIVE STUDY

FEATURE	EXISTING DYSLEXIA APPS	OUR APP
Language Support	Mainly English	Supports English + regional Indian languages
Screening Method	Limited tasks	AI-assisted ,multimodal screening
Engagement	Basic exercises	Gamified tasks, badges, interactive stories
Accessibility	Most not child-friendly	Dyslexia-friendly fonts, color overlays, simple UI
Intervention Plan	Very limited	Full roadmap + personalized tasks + progress tracking
Cost	Many apps paid	Made free/affordable for Indian families
Teacher/Parent Involvement	Minimal	Full dashboards for progress monitoring
Cultural Relevance	Western contexts	Indian bilingual literacy contexts

4.PROBLEM STATEMENT

Many children in India with dyslexia are **not being identified** because the available testing tools are **hard to find**, **too costly**, or **don't match the local languages and culture**. The apps currently used to help these children are broken up, don't give personalized lessons, and struggle with kids who **speak more than one language.** We urgently need one easy-to-use digital program that can spot dyslexia early and offer continuous, interactive help with reading.

5.OBJECTIVES

1. To build an early dyslexia screening app using **AI-assisted adaptive testing.**
2. To provide **bilingual** and **culturally relevant learning** modules for Indian children.
3. To integrate **multisensory literacy-building activities** (audio, visual, tracing).
4. To offer **personalized** intervention roadmaps after screening.
5. To support **parents, teachers, and schools** with **real-time progress tracking.**
6. To **increase awareness** about dyslexia and reduce stigma.

6. PLANNING OF WORK

6.1. Methodology

The project will follow an **Agile and Iterative Development methodology**, integrated with **User-Centric Design (UCD)** principles. This ensures continuous feedback from educational psychologists, parents, and children, enabling rapid prototyping and meaningful refinement.

Key Phases of Methodology

1. **Research & Consultation**
    - Conduct in-depth discussions with educational psychologists and dyslexia specialists.
    - Validate screening metrics and intervention pedagogy.
2. **Prototyping & Wireframing**
    - Create low-fidelity and high-fidelity UI mockups.
    - Implement dyslexia-friendly design elements (custom fonts, high readability overlays, color contrast optimization).
3. **Module Development (Iterative)**
    - Develop the core modules—Screening, Gamified Intervention, Parent Dashboard—through short sprints.
    - Integrate feedback continuously.
4. **Usability Testing**
    - Perform small-scale tests with children and parents.
    - Ensure intuitive navigation, engagement, and accessibility.

5. **Expert Validation & Efficacy Testing**
    - Validate accuracy of the AI-based screening algorithm.
    - Measure impact on literacy skills over a defined period.

6.2. Proposed Solution: The ReadBright Application

ReadBright is a comprehensive, early-stage dyslexia support application offering screening, structured intervention, and continuous family-school integration.

1. System Architecture

The system uses a **multi-layered architecture** to support real-time processing, adaptive AI, and a seamless user experience.

Front-End (User Interface)

- Built using **Flutter (Dart)** or **React Native** for cross-platform support (Android & iOS).
- UI strictly follows **dyslexia-friendly** standards (OpenDyslexic font, simple layouts, controlled animations).

Gaming Modules

- Developed using **Unity (C#)** for interactive 2D/3D learning experiences.
- Includes gamified literacy-building tasks integrated with API calls from the main app.

Back-End (Logic & Data Processing)

- Built using **Python** for adaptive AI logic and analytics.
- **Node.js** handles scalable APIs and real-time data sync.

Databases

- **PostgreSQL (SQL):** For structured data such as student progress, screening results, and profiles.

- **MongoDB (NoSQL):** For flexible storage, dynamic game configurations, logs, and user-generated content.

2. Key Features

The application is structured around **three main pillars: Screening, Intervention, and Support**.

A. AI-Driven Screening Module

- **Adaptive Testing:**
  Difficulty dynamically adjusts based on responses, ensuring accuracy in identifying deficits (phonological awareness, rapid naming, working memory).

- **Multimodal Analysis:**
  Uses speech input, response time, behavioral patterns, and error categories to generate risk profiling.

- **Cultural Relevance:**
  Available in major Indian languages and designed to remove cultural bias.

B. Gamified Intervention Module

- **Multisensory Learning:**
  Activities combine audio cues, visual elements, and kinesthetic actions (drag, tap, trace), based on Orton–Gillingham (O-G) multisensory principles.

- **Scaffolded Learning Path:**
  Starts from phonemic awareness → blending → decoding → fluency → comprehension.

- **Engagement Systems:**
  Includes personalized avatars, reward badges, levels, and progress-based challenges to enhance motivation and confidence.

C. Family and School Integration (Support Module)

- **Parent/Teacher Dashboard:**
  Shows progress reports, skill deficiency breakdowns, and suggested practice modules.

- **Intervention Roadmap:**
  Provides actionable guidance, recommended activities, and optional links to professional resources.

- **Communication Portal:**
  Secure in-app messaging for parents and teachers to coordinate the child's support strategies.

6.3. Project Timeline and Next Steps

The project is distributed across **four semesters**, aligned with B.Tech PBL guidelines.

Semester	Phase Title	Key Activities	Deliverables
3rd Semester	*Concept & Literature Review*	- Initial concept presentation - Literature review - Finalization of problem statement and objectives	PBL-I Synopsis Report (This Document) Initial System Architecture
4th Semester	*Foundation & Design*	- Learn Flutter/Unity - Consult psychologists - Detailed UI/UX design - Begin development (login, user profiles)	Detailed Design Document Functional Prototype (Login + Dashboard)
5th Semester	*Development & Testing*	- Develop core features (Screening + 50% Intervention Modules) - Alpha testing - Usability testing with children	Beta Version Usability Test Reports Refined AI Model
6th Semester	*Final Product & Efficacy*	- Deployment preparation - Efficacy testing - Documentation	Final Application Efficacy Report Project Documentation